SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – DEBUGGING & OPTIMIZATION TASK

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	K.Madhumitha	**Reg. No.:**	192513069
Section / Batch:	BE EEE	**Date:**	24/07/2026

Code of Conduct

I K.Madhumitha [192513069] *(Name / Reg No)* *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K.Madhumitha

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25
Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
		GRAND TOTAL:	**100 Marks**

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well-documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well-justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided

Faculty In-charge	Course Coordinator	Program Director
Signature: ______________	Signature: ______________	Signature: ______________
Date: __________________	Date: __________________	Date: __________________

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5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List
5 marks

Q9
Linked List

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
dequeue(q):
if front==-1:
print("underflow")
    else:
item=q[front]
front=front+1
    if front>rear
    front=-1
        rear=-1
        return item
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19251306
9RA_6a789c17ed5181.96608087\main
.c:1:1: warning: return type
defaults to 'int' [-Wimplicit-
int]
```

Run Save Submit Fix

Search 29°C Partly cloudy ENG IN 20:59 09/08/2026

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Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

Q8

```
printf("underflow");
return -1;
}
return stack[top--];
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int pop()
{
    if(top == -1)
    {
        printf("underflow");
        return -1
    }
    return stack[top--1];
}
```

Test Input

```
10
20
pop
```

Output

Search

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List

Q3 Stack

5 marks

Check the snippet and identify the logical error.

int stack[5];

int top;

void init() {

top = 0;

}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int stack[5];
int top;
void intt()
{
    top = -1
}
```

Test Input

10

Output

Search

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5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack

```
if(ch == ')') {
while(stack[top] != ')') {
postfix[j++] = pop();
}
pop();
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if((ch==')')
{
    while(top!= -1 && stack[top]!= '(')
    {
        postfix[j++]=pop();
    }
    pop();
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19251306
```

Search

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Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

In balancing Bracket, you identify the error.

```
if((ch == ']' && stack[top] == '[') ||
(ch == ')' && stack[top] == '(')) {
pop();
} else {
return 0;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if(ch== ')' && stack[top]== '(')||
    (ch=='|') && stack[top]== '[')||
     (ch== '}' && stack[top]=='{')){
      pop();
} else{
    return 0;
}
```

Test Input

stdin input (optional)

Output

Compilation Error:

Search

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QUESTIONS

Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack - Balancing Symbols 5 marks

Q6 Singly Linked List

5 marks

What is the issue?

```
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
tem=head
while temp.next!=null;
print(temp.data)
    temp=temp.next
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
pop(stack)
    if top==-1:
print("underflow")
    else:
return stack[top--]
```

Test Input

stdin input (optional)

Search

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QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6

Q8 **Linked List** 5 marks

Identify the errors in the following code

while current != NULL:

current.next = prev

prev = current

current = current.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
while current!=null
next=current.next
current.next=prev
prev=current
current=next
```

Test Input

stdin input (optional)

Output

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1)

← Back

QUESTIONS

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols

Q9 **Linked List**

5 marks

What is the error in the below code?

new.next = head

head = new

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
new=node(data)
    new.next=head
    head=new
```

Test Input

stdin input (optional)

Output

Search

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Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack - Balancing Symbols
5 marks

Q6
Singly Linked List
5 marks

Q7
Stack
5 marks

```
if front == -1:
print("Underflow")
else:
return Q[front++]
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
dequeue(q):
if front==-1:
print("underflow")
    else:
item=q[front]
front=front+1
    if front>rear
    front=-1
        rear=-1
        return item
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19251306
9RA_6a735e17ed5101.96608007\main
.c:1:1: warning: return type
defaults to 'int' [-Wimplicit-
```

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List

Q1 **Stack** 5 marks

Point out the error?
PEEK(stack):
return stack[top+1]

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
peek(stack):
return stack[top]
```

Test Input

stdin input (optional)

Output

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if((rear+1)% max==front){
    printf("queue full");
}
else:
rear=(rear+1) % max ;
queue[rear]=vehicle;
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19251306
9RA_6a7981f05f9952.26274509\main
.c:1:1: error: expected
identifier or '(' before 'if'
```

Search

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QUESTIONS

Q1 Stack 5 marks

Q2 Circular Queue 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Doubly Linked List

Q3 Stack

5 marks

Point out the error.

if top == MAX:

Overflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
if top==max-1:
print("overflow")
```

Test Input

stdin input (optional)

Output

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QUESTIONS

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List

Q4 **Stack** 5 marks

Identify the issue in the below pseudocode.
POP():
if top == 0 → Underflow

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
pop():
if top==-1
```

Test Input

stdin input (optional)

Output

Search

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

```
del->next->prev = del->prev;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void deletenode(node*del){
    if (del->prev!=null)
        del->prev->next=del->next;
    else
        head=del->next;
    if(del->next!=null)
        del->next->prev=del->prev;
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19251306
9RA_6a7982db08cbd3.50495305\main
.c:1:17: error: unknown type
```

Q1
Stack
5 marks

Q2
Circular Queue
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {
if (rear == MAX) return;
queue[rear++] = data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void enqueue(int data)
{
    if(rear==max-1) return;
    queue[rear++]=data;
}
```

Test Input

stdin input (optional)

Output

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
void insertfront(int data){
    if(front==-1){
        front=rear=0;
    }
    else if(front==0){
        front=max-1;
    }
    else{
        front=front-1;
    }
    deque[front]=data;
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19251306
9RA_6a7903a8627ba5.18602957\main
.c: In function 'insertfront':
C:\Windows\TEMP\sandbox_19251306
```

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

Q8

}

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int binarysearch(int arr[],int 1,int r,int x)
{
    if(1>r) return-1;
    int mid=1+(r-1)/2;
    if(arr[mid]==x) return mid;
    if (arr[mid]>x)
        return binarysearch(arr,1,mid-1,x);
    return binarysearch(arr,mid+1,r,x);
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19251306
9RA_6a7984948efc49.49731620\main
.c:1:32: error: expected ';',
',' or ')' before numeric
constant
```

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Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

```
return data;
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int dequeue(){
    if(front==-1) return-1;
    int data=queue[front];
    if(front=rear)
        front=rear=-1;
    else
        front=(front+1)% max;
    return data;
}
```

Test Input

stdin input (optional)

Output

```
Compilation Error:
C:\Windows\TEMP\sandbox_19251306
9RA_6a798560957b26.58530143\main
.c: In function 'dequeue':
```

Search

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5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Doubly Linked List
5 marks

Q6
Queue
5 marks

Q7
Queue - Deque
5 marks

```
int op1 = pop();
int op2 = pop();
// perform operation: result = op1 + op2;
push(result);
}
}
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
int op1=pop();
int op2=pop();
result=op2 - op1; //for substraction
push(result);
```

Test Input

stdin input (optional)

Output

Search

35°C Sunny

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